MARK SCHEME · TEACHER COPY

10BR · Chapter 2: Data Transmission

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Total marks: 75

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

For teacher / admin use only. Do not give to students before they have attempted the paper.

Mark scheme extracts © Cambridge University Press & Assessment.

Question 1 · Oct/Nov 2023 Paper 11, original Q7 [4 marks]

Question	Answer	Marks
1	**One** mark for each correct term, in the correct order: – header – destination address – routers – last	4

Question 2 · Feb/March 2024 Paper 12, original Q7 (a),(b) [8 marks]

Question	Answer	Marks
2(a)(i) i)	If the data is intercepted, it cannot be understood	1
2(a)(ii)	**Four** from: • Symmetric has a shared key… • … to encrypt **and** decrypt • Both the sender and receiver know the key • Asymmetric has different keys // a public key **and** a private key • …public to encrypt the data **and** private to decrypt • …anyone can know the public key but only those intended know the private key	4
2(b)(i)	Any **two** from: e.g. • Destination address/IP • Sender address/IP • Packet number	2
2(b)(ii)	Any **one** from: • Control the route the packet takes • Send each packet towards its destination • Choose more efficient route	1

Total: 75 marks

Question 3 · May/June 2024 Paper 11, original Q4 (a),(b),(c) [7 marks]

Question	Answer	Marks
3(a)	Any **three** from: • A packet is split into **three** different sections • … the header • … the payload • … the trailer	3
3(b)	Router	1
3(c)	Any **three** from: • The network may be spread over a long distance … • … so it is more **reliable** • Bits will be sent/arrive in **sequence** • …so bits less likely to be skewed • Less crosstalk/interference • … so less likely to have errors • The data may not need to be transmitted at a fast speed // data transmission speed of serial is adequate • The cables in the network only use serial transmission	3

Question 4 · May/June 2024 Paper 12, original Q6 [5 marks]

Question	Answer	Marks
4	**One** mark for each correct method.	5

error detection method	statement
parity (check/bit/byte/block)	An odd or even process can be used.
checksum	A value is calculated from the data using an algorithm. This happens before and after the data is transmitted.
echo check	A copy of the data is sent back to the sender by the receiver.

automatic repeat query/request // ARQ	Acknowledgement and timeout are used.
check digit	A value is appended to data that has been calculated using the data. This value is checked on data entry.

Question 5 · Oct/Nov 2024 Paper 11, original Q4 (a),(b) [11 marks]

Question	Answer	Marks
5(a)	Any **four** from: • The data packet has three sections • It has a packet header … • … that contains data such as the destination address • It has a payload … • … that contains the main data for the email • It has a trailer … • … that contains data such as the error detection system used	**4**
5(b)(i)	Any **four** from: • It sends the data multiple bits at the same time // It uses multiple wires • … so the transmission speed of the data will be fast • Data may not need to travel a long distance • … as the devices are all within a single room • It sends data in both directions at the **same time** • … so users on the network can send data to each other with **no delay**	**4**
5(b)(ii)	Any **two** from: • More interference/crosstalk (due to multiple wires) • **Data** may be skewed (due to multiple bits at a time) // bits may arrive out of order • More chance of data collisions (as data sent in both directions at the same time) • More chance of error in the data	**2**

5(b)(iii)	Any **one** from: • Serial simplex • Serial half-duplex • Serial full-duplex • Parallel simplex • Parallel half-duplex	**1**

Question 6 · May/June 2025 Paper 11, original Q4 (a),(b),(c) [12 marks]

Question	Answer	Marks
6(a)(i)	**Two** from (one mark for serial, one mark for full-duplex): **serial** • Serial is suitable as data may be travelling a long distance (over 5 metres). • Serial is suitable to make sure that bits don't arrive skewed/arrives in order. • Serial is suitable as there is less chance of error. • Serial has sufficient transmission speed as not much data being sent. **full-duplex** • Full-duplex is suitable as data needs to go between customer and kitchen computer // data needs to go in both directions // so notification of errors can be sent back to customer.	**2**
6(a)(ii)	The data may be transmitted faster.	**1**
6(b)(i)	Any **five** from: • The checksum is **calculated** from/using the **data**. • … using an algorithm // by example. • The (checksum) value is transmitted with the data. • The (checksum) value is recalculated **after transmission/by recipient**. • If the **checksum values** do not match, an error is detected // If the **checksum values** match, no error is detected … • … request for data to be resent if there is an error.	**5**
6(b)(ii)	Any **one** from: • Parity byte/block check • Echo check	**1**

6(c)(i)	D	1
6(c)(ii)	Any **Two** from: • The packets **arrive** in the wrong order. • Each packet could take a different route. • Some packets may take longer than others to be transmitted. • Some packets may not arrive/be corrupted and need to be requested again.	2

Question 7 · May/June 2025 Paper 13, original Q7 (a),(b) [10 marks]

Question	Answer	Marks
7(a)	• It can suffer Interference/crosstalk. • … that could have caused data loss/gained/changed.	2
7(b)	**One** mark for each correct term in the correct place. • odd/even • even/odd (must be opposite of MP1) • bit • echo check • automatic repeat query (ARQ) • timeout • positive/negative or negative/positive A parity check is set to be **odd** or **even**. A parity **bit** is added to each byte before transmission to make it match the set parity. An **echo check** involves comparing the data sent to the data received back from the receiving device. An **automatic repeat query (ARQ)** uses acknowledgement and **timeout**. The acknowledgement system can be **positive** or **negative**.	8

Total: 75 marks

Question 8 · Oct/Nov 2025 Paper 13, original Q3 (a),(b) [14 marks]

Question	Answer	Marks
8(a)	**One** mark for each correct section **One** mark for each correct example of what it stores First section • Header • Originators/senders (IP) address • Destination/receivers (IP) address • Packet number Second section • Payload • The data itself Third section • Trailer • End of packet notification • Error checking data	**6**

Total: 75 marks

8(b)(i)	**Four** from: • Data is sent one <u>bit</u> at a time • … down a single wire • Data can be sent in both directions … • … at the same time	**4**
8(b)(ii)	Any **Four** from: • Data is sent multiple <u>bits</u> at the **same time** • … down multiple wires • Data can be sent in both directions … • … not at the same time	**4**

Question 9 · May/June 2026 Paper 11, original Q4 (b) [4 marks]

Question	Answer	Marks	Guidance
9 ·	Examiners must **consistently** apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.		
9(a)	**Four** from: • (Serial) transmits the data one bit at a time • … down a single wire/channel/line • (Half-duplex) transmits the data (between the computer and the mouse) in both directions/ways … • … but only one direction/way at a time	**4**	MP1 – BOD bit by bit MP2 –cable/path is TV